

# Condition Number and Matrix Norms

## Numerical Analysis

**Zamir Hussain**

Department of Mathematics  
University of Wah

February 10, 2026

---

## 1 Introduction: The Stability of Linear Systems

In Numerical Analysis, when solving a system of linear equations represented by  $AX = b$ , it is crucial to determine if the system is **stable**. We often deal with approximations and round-off errors. A fundamental question arises:

*Does a small error in the input data result in a small or large error in the final solution?*

### 1.1 Classification of Systems

Based on how systems respond to perturbations (small changes) in  $A$  or  $b$ :

- › **Well-Conditioned:** A small change in input ( $A$  or  $b$ ) results in a **small** change in the solution.
- › **Ill-Conditioned:** A small change in input results in a **drastic** change in the solution.

#### **Stability Warning**

##### **Illustrative Example of Instability:**

Consider a system where the exact solution is  $X = (1, 1)$ . If a change of 0.1 in the vector  $b$  results in a solution like  $X = (82, -15)$ , the system is considered unstable or ill-conditioned.

## 2 The Condition Number

To quantify *how* ill-conditioned a system is, we use a metric called the **Condition Number**.

## 2.1 Mathematical Derivation

Let  $AX = b$  be the original system. Let us perturb  $b$  by a small amount  $\delta b$ , resulting in a change in the solution  $\delta X$ .

$$\begin{aligned} A(X + \delta X) &= b + \delta b \\ AX + A(\delta X) &= b + \delta b \\ \implies A(\delta X) &= \delta b \quad (\text{Since } AX = b) \\ \delta X &= A^{-1} \delta b \end{aligned} \tag{1}$$

Taking the norm on both sides and using the property  $\|AB\| \leq \|A\|\|B\|$ :

$$\|\delta X\| \leq \|A^{-1}\| \|\delta b\| \tag{2}$$

From the original system  $\|b\| \leq \|A\|\|X\|$ , which implies  $\frac{1}{\|X\|} \leq \frac{\|A\|}{\|b\|}$ . Combining these inequalities, we get the relationship for Relative Error:

$$\frac{\|\delta X\|}{\|X\|} \leq \underbrace{\|A\| \cdot \|A^{-1}\|}_{\text{Condition Number}} \cdot \frac{\|\delta b\|}{\|b\|} \tag{3}$$

### Definition: Condition Number

For any invertible matrix  $A$ , the Condition Number, denoted as  $K(A)$  or  $\text{cond}(A)$ , is defined as:

$$K(A) = \|A\| \cdot \|A^{-1}\| \tag{4}$$

- If  $K(A) \approx 1$ , the matrix is **Well-Conditioned**.
- If  $K(A) \gg 1$  (Very Large), the matrix is **Ill-Conditioned**.

## 3 Calculating the Condition Number

The value of  $K(A)$  depends on the specific matrix norm used.

### 3.1 Maximum Row Sum Norm ( $l_\infty$ )

The infinity norm is the maximum absolute row sum:

$$\|A\|_\infty = \max_i \sum_{j=1}^n |a_{ij}| \tag{5}$$

### 3.2 Spectral Norm ( $l_2$ )

This norm uses eigenvalues. If  $A$  is **Real Symmetric** or **Hermitian**, the formula simplifies significantly:

$$K(A)_2 = \frac{|\lambda_{\max}|}{|\lambda_{\min}|} \tag{6}$$

Where  $\lambda$  represents the eigenvalues of  $A$ .

## 4 Solved Examples

### Example 1: Using Row Sum Norm ( $l_\infty$ )

Find the condition number of matrix  $A$  using the infinity norm.

$$A = \begin{bmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{bmatrix}, \quad A^{-1} = \frac{1}{8} \begin{bmatrix} -31 & 44 & -17 \\ 44 & -56 & 20 \\ -17 & 20 & -7 \end{bmatrix}$$

**Solution:**

**Step 1: Calculate  $\|A\|_\infty$**

- Row 1:  $|1| + |4| + |9| = 14$
- Row 2:  $|4| + |9| + |16| = 29$
- Row 3:  $|9| + |16| + |25| = 50$

Thus,  $\|A\|_\infty = 50$ .

**Step 2: Calculate  $\|A^{-1}\|_\infty$**  Ignore the  $1/8$  factor momentarily:

- Row 1:  $|-31| + |44| + |-17| = 92$
- Row 2:  $|44| + |-56| + |20| = 120$
- Row 3:  $|-17| + |20| + |-7| = 44$

Max sum is 120. Apply factor:  $\|A^{-1}\|_\infty = \frac{120}{8} = 15$ .

**Step 3: Condition Number**

$$K(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty = 50 \times 15 = \mathbf{750}$$

*Conclusion: The system is Ill-Conditioned.*

### Example 2: Spectral Condition Number

Determine  $K(A)_2$  for the matrix from Example 1, given its eigenvalues are  $\lambda \approx 37.5, -2.58, 0.0825$ . Note that  $A$  is symmetric.

**Solution:** Since  $A$  is symmetric, we define:

$$K(A) = \frac{|\lambda_{\max}|}{|\lambda_{\min}|}$$

$$|\lambda_{\max}| = 37.5, \quad |\lambda_{\min}| = 0.0825$$

$$K(A) = \frac{37.5}{0.0825} \approx \mathbf{454.5}$$

### Example 3: Optimization

Let  $A(\alpha) = \begin{bmatrix} 0.1\alpha & 0.1\alpha \\ 1.0 & 1.5 \end{bmatrix}$ . Minimize  $K(A)_\infty$ .

**Solution:**

- $\|A\|_\infty = \max(0.2|\alpha|, 2.5)$ .

- Calculate  $A^{-1}$ : Determinant  $= 0.05\alpha$ .

$$A^{-1} = \frac{1}{0.05\alpha} \begin{bmatrix} 1.5 & -0.1\alpha \\ -1 & 0.1\alpha \end{bmatrix} = \frac{1}{|\alpha|} \begin{bmatrix} 30 & -2\alpha \\ -20 & 2\alpha \end{bmatrix}$$

- $\|A^{-1}\|_{\infty} = \frac{1}{|\alpha|}(30 + 2|\alpha|) = \frac{30}{|\alpha|} + 2$ .

To minimize the product, find where the max terms intersect:

$$0.2|\alpha| = 2.5 \implies |\alpha| = 12.5$$

Therefore,  $\alpha = 12.5$  **or**  $-12.5$ .

## 5 Upper Bound on Relative Error

Sometimes we have an approximate solution  $X^*$  and define the **residual** vector  $r = b - AX^*$ .

### Theorem: Error Bound using Residuals

For an approximate solution  $X^*$ , the relative error is bounded by:

$$\frac{\|X - X^*\|}{\|X\|} \leq K(A) \frac{\|r\|}{\|b\|} \quad (7)$$

## 6 Summary

- **Stability:** Numerical solutions are stable if small input errors yield small output errors.
- **Condition Number:**  $K(A) = \|A\| \cdot \|A^{-1}\|$  measures the amplification of error.
- The Condition Number is dependent on the type of norm used (Spectral vs Infinity).
- If working with precision  $10^{-k}$  and  $K(A) = 10^m$ , you likely lose  $m$  digits of accuracy in the final result.

## Practice Problems

1. **Condition Number Calculation:** Given  $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ .
  - (a) Find  $A^{-1}$ .
  - (b) Calculate  $\|A\|_\infty$  and  $\|A^{-1}\|_\infty$ .
  - (c) Compute  $K(A)$ . Is it well-conditioned?
2. **Spectral Condition Number:** Find the condition number of the diagonal matrix  $D = \text{diag}(100, 2, 0.1)$  using the Spectral (2-norm) method.
3. **Conceptual:** If  $K(A) = 10^6$  and you are working with a computer that has 10 decimal digits of precision, how many accurate digits can you trust in your solution?

## 7 Geometry and Sensitivity

Numerical conditioning is deeply rooted in geometry. Understanding the shape of linear transformations provides intuition for why some systems are sensitive to errors while others are not.

### 7.1 Geometric Interpretation of $Ax = b$

Consider the mapping  $x \mapsto Ax$ .

- **Unit Sphere to Ellipsoid:** If we take all vectors  $x$  on the unit circle (or sphere in higher dimensions), i.e.,  $\|x\|_2 = 1$ , the matrix  $A$  transforms this shape into a hyper-ellipsoid.
- **Singular Values ( $\sigma$ ):** The lengths of the principal axes of this ellipsoid are given by the singular values of  $A$ , denoted  $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n > 0$ .

#### Condition Number via SVD

In the Euclidean norm ( $l_2$ ), the condition number is the ratio of the largest to the smallest stretching factor:

$$K(A)_2 = \frac{\sigma_{\max}}{\sigma_{\min}} \quad (8)$$

### 7.2 Visualizing Conditioning

1. **Well-Conditioned ( $K \approx 1$ ):** The ellipsoid is nearly spherical. The mapping rotates or scales space evenly. A small error ball in the input remains a small error ball in the output.
2. **Ill-Conditioned ( $K \gg 1$ ):** The ellipsoid is extremely thin (needle-shaped or pancake-shaped).
  - In a 2D system (intersection of two lines), ill-conditioning means the lines are **nearly parallel**.
  - Finding the intersection point of nearly parallel lines is numerically unstable; a tiny shift in one line sends the intersection point infinitely far away.

## 8 ML Applications: Stability and Robustness

In Machine Learning and Data Science, matrix operations are ubiquitous. The condition number plays a critical role in the stability of training algorithms.

### 8.1 Linear Regression and Normal Equations

In least squares regression, we find weights  $\theta$  by solving the normal equations:

$$(X^T X)\theta = X^T y \quad (9)$$

If the features in dataset  $X$  are highly correlated (multicollinearity), the matrix  $A = X^T X$  becomes nearly singular (very high condition number).

### ⚠ Stability Warning

**Consequence in ML:** If  $X^T X$  is ill-conditioned, the learned weights  $\theta$  become unstable. They may become massively large or oscillate wildly due to minor noise in the training labels  $y$ .

## 8.2 Robustness via Regularization

To fix an ill-conditioned system, we often apply **Tikhonov Regularization** (Ridge Regression in ML). We modify the equation to:

$$(X^T X + \lambda I)\theta = X^T y \quad (10)$$

**Why this works numerically:** Adding  $\lambda I$  shifts the eigenvalues. If the eigenvalues of  $X^T X$  were roughly  $\{100, 0.001\}$ , the condition number was  $10^5$ . Adding  $\lambda = 0.1$ :

$$\text{New } \lambda \approx \{100.1, 0.101\} \implies K \approx \frac{100.1}{0.101} \approx 1000$$

The system becomes drastically more stable.

## 8.3 Deep Learning Gradients

In Deep Neural Networks, gradients are computed via backpropagation, which involves successive matrix multiplications.

- **Exploding Gradients:** If spectral norms (singular values) of weight matrices are  $> 1$ , gradients grow exponentially.
- **Vanishing Gradients:** If spectral norms are  $< 1$ , gradients decay to zero.
- **Solution:** Weight initialization techniques (like Xavier/Glorot) aim to keep the initial singular values close to 1 (well-conditioned).

## 9 Worked Examples

### 9.1 Example 1: Condition Number Calculation

#### Condition Number of a Perturbed Matrix

Compute the condition number of:

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 2.001 \end{bmatrix}$$

**Solution:**

$$\det(A) = 1 \times 2.001 - 2 \times 1 = 0.001$$

$$A^{-1} = \frac{1}{0.001} \begin{bmatrix} 2.001 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2001 & -2000 \\ -1000 & 1000 \end{bmatrix}$$

$$\|A\|_{\infty} = \max(1 + 2, 1 + 2.001) = 3.001$$

$$\|A^{-1}\|_{\infty} = \max(|2001| + |-2000|, |-1000| + |1000|) = 4001$$

$$\kappa_{\infty}(A) = 3.001 \times 4001 \approx \mathbf{12007}$$

*Conclusion:* A large condition number indicates high sensitivity.

### 9.2 Example 2: Sensitivity Analysis

#### Sensitivity Analysis

Consider  $A = \begin{bmatrix} 1 & 1 \\ 1 & 1.001 \end{bmatrix}$ ,  $\mathbf{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ . Exact solution:  $\mathbf{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$ .

If  $\mathbf{b}$  changes to  $\begin{bmatrix} 2 \\ 2.001 \end{bmatrix}$ , find the error bound.

**Solution:**

$$\Delta \mathbf{b} = \begin{bmatrix} 0 \\ 0.001 \end{bmatrix}$$

$$\frac{\|\Delta \mathbf{b}\|}{\|\mathbf{b}\|} = \frac{0.001}{2} = 0.0005 \quad (\text{using } l_{\infty})$$

$$\kappa(A) \approx 4000 \quad (\text{Derived similarly to Ex 1})$$

$$\text{Expected Relative Error} \leq \kappa(A) \cdot \text{Rel Error in Input}$$

$$\leq 4000 \times 0.0005 = 2 \quad (200\% \text{ error potential})$$

*Check:* The new exact solution is  $x_{\text{new}} = [1, 1]^T$ . The error is massive compared to original  $x = [2, 0]^T$ .



### 9.3 Example 3: The Hilbert Matrix

#### The Hilbert Matrix

Analyze the  $4 \times 4$  Hilbert matrix:

$$H_4 = \begin{bmatrix} 1 & 1/2 & 1/3 & 1/4 \\ 1/2 & 1/3 & 1/4 & 1/5 \\ 1/3 & 1/4 & 1/5 & 1/6 \\ 1/4 & 1/5 & 1/6 & 1/7 \end{bmatrix}$$

#### Facts:

- $\det(H_4) \approx 1.65 \times 10^{-7}$
- $\kappa_2(H_4) \approx 1.55 \times 10^4$
- Already moderately ill-conditioned at size 4.
- $\kappa(H_{10}) \approx 1.6 \times 10^{13}$  (extremely ill-conditioned; notoriously difficult to solve numerically).

## 10 Exercises

### 10.1 Problem 1: Norm Calculations

- (a) Compute  $\|\mathbf{x}\|_1$ ,  $\|\mathbf{x}\|_2$ ,  $\|\mathbf{x}\|_\infty$  for  $\mathbf{x} = [3, -4, 12]^T$ .
- (b) For  $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$ , compute  $\|A\|_1$ ,  $\|A\|_\infty$ ,  $\|A\|_F$  (Frobenius).

### 10.2 Problem 2: Condition Number Analysis

- (a) Compute  $\kappa_2(A)$  for  $A = \begin{bmatrix} 10 & 7 \\ 7 & 5 \end{bmatrix}$ .
- (b) Show that for any orthogonal matrix  $Q$ ,  $\kappa_2(Q) = 1$ .
- (c) Prove that  $\kappa(A) \geq 1$  for any non-singular matrix  $A$ .

### 10.3 Problem 3: Sensitivity Analysis

Solve  $A\mathbf{x} = \mathbf{b}$  where:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Then perturb to  $\mathbf{b}' = \begin{bmatrix} 2 \\ 2.0001 \end{bmatrix}$  and solve again. Compare the relative change in solution to the bound provided by the condition number.

### 10.4 Problem 4: Geometric Interpretation

Consider the transformation by matrix:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 0.01 \end{bmatrix}$$

- (a) What is the condition number  $\kappa_2(A)$ ?
- (b) Describe what happens to the unit circle under this transformation.
- (c) What does this imply about the stability of solving  $A\mathbf{x} = \mathbf{b}$ ?

### 10.5 Problem 5: Regularization (ML)

Given the ill-conditioned system from Problem 3, Apply Tikhonov regularization with  $\lambda = 0.01$ . Compute  $(A^T A + \lambda I)$  and check if the condition number improves compared to  $A^T A$ .

## 11 Solutions to Selected Exercises

### 11.1 Solution to Problem 1

(a) Vector norms:

$$\|\mathbf{x}\|_1 = |3| + |-4| + |12| = 3 + 4 + 12 = \mathbf{19}$$

$$\|\mathbf{x}\|_2 = \sqrt{3^2 + (-4)^2 + 12^2} = \sqrt{9 + 16 + 144} = \sqrt{169} = \mathbf{13}$$

$$\|\mathbf{x}\|_\infty = \max(|3|, |-4|, |12|) = \mathbf{12}$$

(b) Matrix norms:

$$\|A\|_1(\text{max col sum}) = \max(|2| + |3|, |-1| + |4|) = \max(5, 5) = \mathbf{5}$$

$$\|A\|_\infty(\text{max row sum}) = \max(|2| + |-1|, |3| + |4|) = \max(3, 7) = \mathbf{7}$$

$$\|A\|_F = \sqrt{2^2 + (-1)^2 + 3^2 + 4^2} = \sqrt{4 + 1 + 9 + 16} = \sqrt{30} \approx \mathbf{5.477}$$

### 11.2 Solution to Problem 2

(a) For  $A = \begin{bmatrix} 10 & 7 \\ 7 & 5 \end{bmatrix}$ :

$$\text{Eigenvalues of } A^T A = \text{Eigenvalues of } \begin{bmatrix} 149 & 105 \\ 105 & 74 \end{bmatrix}$$

$$\det(A^T A - \lambda I) = \lambda^2 - 223\lambda + 1 = 0$$

$$\lambda \approx 222.9955 \text{ and } 0.0045$$

$$\sigma_{\max} = \sqrt{222.9955} \approx 14.933, \quad \sigma_{\min} = \sqrt{0.0045} \approx 0.0671$$

$$\kappa_2(A) = \frac{14.933}{0.0671} \approx \mathbf{222.5}$$

(b) For orthogonal  $Q$ ,  $Q^T Q = I$ . Since eigenvalues of  $I$  are all 1, all singular values  $\sigma_i = \sqrt{1} = 1$ . Thus  $\kappa_2(Q) = 1/1 = 1$ .

(c) Proof:  $\kappa(A) = \|A\| \|A^{-1}\|$ . Since  $AA^{-1} = I$  and  $\|I\| \geq 1$  (for consistency norms),  $\|A\| \|A^{-1}\| \geq \|I\| = 1$ .

### 11.3 Solution to Problem 3

**Exact solution:** Solving  $\begin{bmatrix} 1 & 1 \\ 1 & 1.0001 \end{bmatrix} x = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$  yields  $\mathbf{x} = [2, 0]^T$ .

**Perturbed system:** Solving with  $\mathbf{b}' = [2, 2.0001]^T$  yields  $\mathbf{x}' = [1, 1]^T$ .

**Relative change in  $\mathbf{x}$ :**  $\|\Delta \mathbf{x}\| / \|\mathbf{x}\| = \|[1, -1]\| / 2 = 1/2$  (Infinity norm) or  $\sqrt{2}/2 \approx 0.707$  (2-norm).

**Condition Number:**  $\kappa(A) \approx 40000$ .

**Bound:**  $\|\Delta \mathbf{x}\| / \|\mathbf{x}\| \leq \kappa(A) \|\Delta \mathbf{b}\| / \|\mathbf{b}\| \approx 40000 \times \frac{0.0001}{2} = 2$ .

The actual error (0.5 or 0.707) is indeed within the predicted bound (2).